

Tariq Islam

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Summary

Sensor intelligence / DSP + ML engineer building physiology-grounded algorithms for noisy wearable biosignals (PPG/EEG), biomedical images; experienced in filters/FFT, artifact handling, and production-minded evaluation.

Education

Princeton University

PhD in Electrical and Computer Engineering

May 2024

Princeton, NJ

Bangladesh University of Engineering and Technology

MSc in Electrical and Electronic Engineering

Jul 2018

Dhaka, Bangladesh

Bangladesh University of Engineering and Technology

BSc in Electrical and Electronic Engineering

Jun 2016

Dhaka, Bangladesh

Experience

Massachusetts Institute of Technology

Postdoctoral Associate, MIT-Novo Nordisk AI Fellow

Jun 2024 – Present

Cambridge, MA

- Built an EEG biometric identification pipeline, achieving **80% acc. on 109 subjects** across 2+ sessions.
- Co-led research on neural spike prediction from **patch clamp time series** and achieved **avg. 3.7 ms early warning**.
- Led microscopy image segmentation: created a dataset and trained specialized models, improving **Dice from 0.25 to 0.78 (+0.53)** vs. Segment Anything on microscopy benchmarks.
- Designed a topological alignment metric to compare representations and support **model explainability**.
- Lecturer**: MAS.745 (Next-Gen Nano & Biotech, Fall 2025), delivered 12 lectures to 16 students on AI and biotech.

Princeton University

Graduate Research Assistant, William G. Bowen Fellow

Sep 2018 – May 2024

Princeton, NJ

- Built a chest X-ray pipeline to detect outliers for data quality control using DL methods; flagged **~300 corrupted/mislabeled** images across CheXpert, ChestX-ray14, and MURA.
- Built a DL-based X-ray analysis system to discover **disease phenotypes** from COVID-19 by comparing them to prior disease (e.g., normal vs. pneumonia).
- Analyzed optimization dynamics of t-SNE/UMAP; built diagnostics to detect distortions/failure modes and reduced **Force Ratio (distortion metric) by 50%** on handwritten digits, X-rays, and clinical data.

Bangladesh University of Engineering and Technology (BUET)

Researcher and Lecturer

Sep 2016 – Aug 2018

Dhaka, Bangladesh

- Developed wearable PPG heart-rate (HR) pipeline using **adaptive filtering + FFT/spectral analysis with artifact/noise handling**; improved HR accuracy 9% and runtime 83%; built evaluation scripts for repeated-session data.
- Built methods for image **denoising under mixed noise**, improving performance by **2.7% (PSNR)** over prior methods.
- Led projects in biosignal/vision analytics, including disease detection from retinal OCT scans; blood cell detection and counting from smear images.
- Lecturer**: Communication, DSP, Numerical Methods, Measurement & Instrumentation; mentored student projects.

Semion Inc.

Computer Vision Researcher (part-time)

May 2016 – Mar 2018

Dhaka, Bangladesh

- Improved chest X-ray disease detection by **+0.04 AUC** by developing an ensemble DL pipeline; built an evaluation workflow and deployed for practicing doctors.
- Developed NLP document classification models for a customer use-case, reducing manual review by **~30%**.

Technical Skills

Languages: Python, MATLAB, C/C++, SQL

Signal Processing: filtering (FIR/IIR), FFT/spectral analysis, peak detection, windowing, resampling, artifact detection/mitigation, signal quality metrics

ML/Data: PyTorch, scikit-learn, TensorFlow, NumPy, SciPy, Pandas

Time Series: feature engineering, classical ML + deep learning for physiological signals, evaluation/validation

Others: Github